ACDC Power Module

BA100-10000 Series

User Manual



Beian New Energy Co., Ltd.

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1 Scope

The BA100-10000 series high-frequency switching ACDC module is AC input and DC output converter. The ACDC module with adjustable DC output. The module using DSP digital control, resonant soft-switching, active PFC technology, and has multiple patent protections; the module features high power density, high power factor, low harmonics, high efficiency, etc., and has the capability of multiple modules being connected in parallel.

2 Functions

- **High Efficiency**

Reduces energy consumption, meets energy-saving and emission-reduction requirements, and helps customers save on electricity costs.

- **High Power Density**

Saves space for customers and lowers system costs.

- **DSP Digital Control**

Fewer components, higher environmental stability, greater reliability, and easier capacity expansion.

- **Low Input Harmonic Distortion**

Reduces grid pollution and enhances grid adaptability.

- **Wide Input and Output Voltage Range**

Suitable for most applications with varying input and output voltage requirements.

- **Wide Operating Temperature Range**

Designed to withstand most harsh working environments.

- **Hot Swap**

It is convenient to use and can reduce maintenance costs.

- **Comprehensive Fault Self-Detection and Alerts**

Extensive fault detection capabilities for easier customer maintenance.

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3 Parameters

Item	Parameters		
Model	**BA48-9000**	**BA100-10000**	**BA150-10000**
DC Output			
Rated Output Power	9kW	10kW	
Rated Output Current	150A	100A	75A
Output Voltage Range	20VDC ~ 60VDC	30VDC ~ 100VDC	50VDC ~ 150VDC
Current Limited Adjust Range	10%~110% output current adjusted by CANBUS		
Voltage Accuracy	≤± 0.5%		
Current Accuracy	≤ 1%		
Efficiency	≥95%		
AC Input			
Rated Input Power	AC380Vac/480Vac		
Input Voltage Range	323Vac ~ 520Vac(Full Power Output) 260Vac ~ 323Vac(Power Linear Reduction)		
Rated Input Current	< 20A		
Frequency	45Hz~65Hz		
PF	≥0.99		
THD	≤5%		
Protection	Fuse		
Working Condition			
Working ambient temperature	-40~50℃ / 50~75℃(Reduced output)		
Storage temperature	-40℃ ~ 75℃		
Relative humidity	0 ~ 95%		
Altitude	<2000m(Full Power Output)		
Product safety and reliability			
Insulating Strength	The input can withstand 50Hz RMS 2500VAC (or equivalent 3535VDC) pressure for 1 minute. The housing can also withstand this level of voltage for 1 minute.		
	The input can withstand an output voltage of 3000VAC (or equivalent 4242VDC) at an effective value of 50Hz for 1 minute.		
MTBF	> 120000 hours		
Dimension/Weight			
Dimension	W130mm * H84mm * L396mm		
Net Weight	7kg		
Communication and Warning			
Interface	CAN		
Module Max Parallel Qty	60 Pcs		

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The alarms and status are reported to the monitor via the CAN communication port, and are also indicated by three panel LEDs.

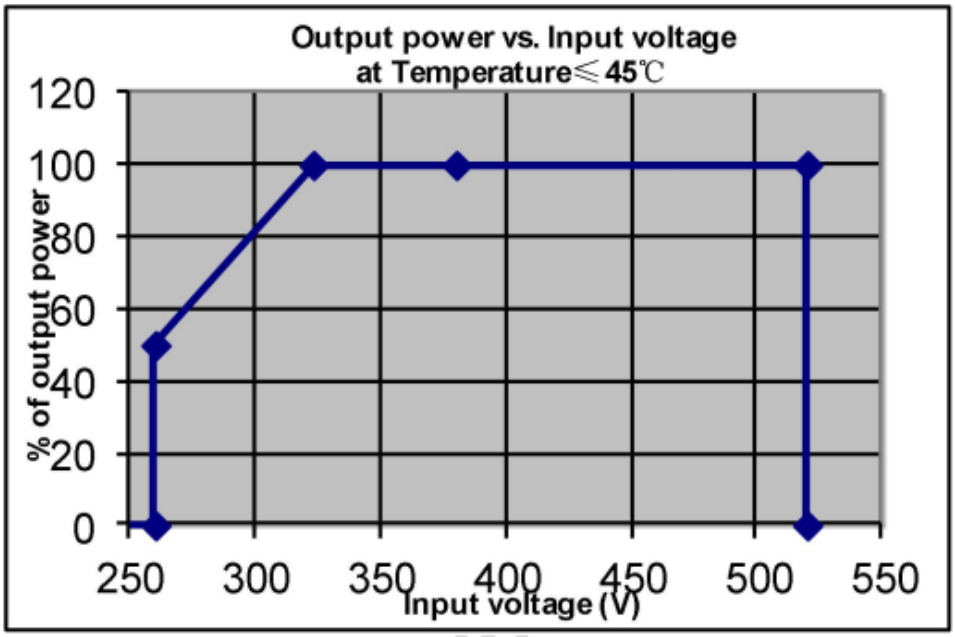


Figure 1.1 Input Limit Power Curve of BA100-10000 Series

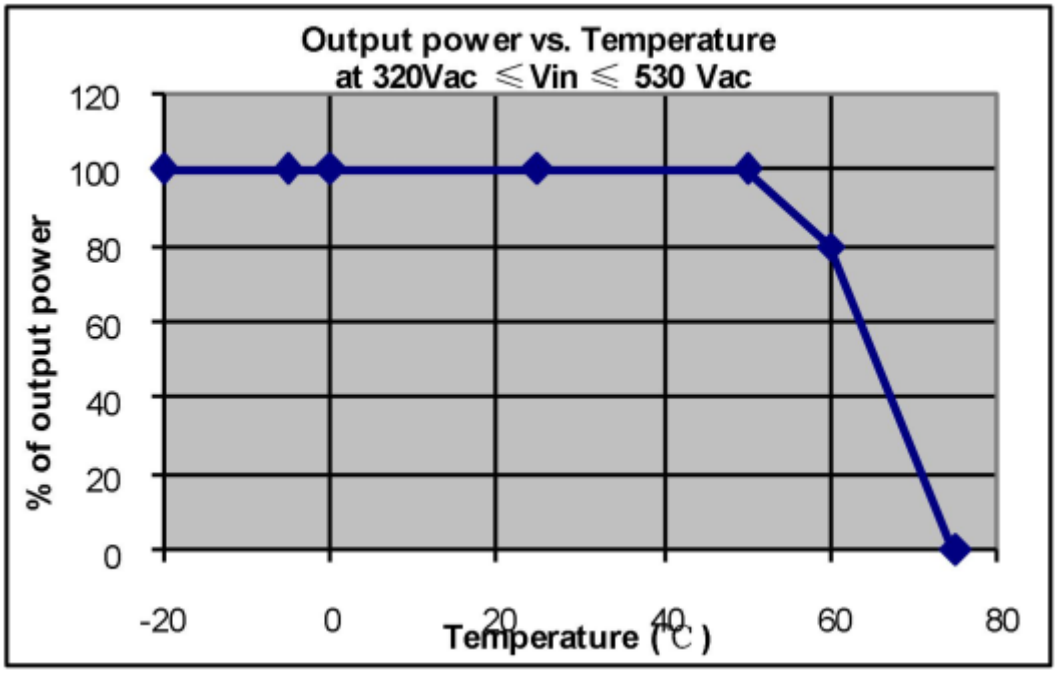


Figure 1.2 Temperature Limit Power Curve of BA100-10000 Series

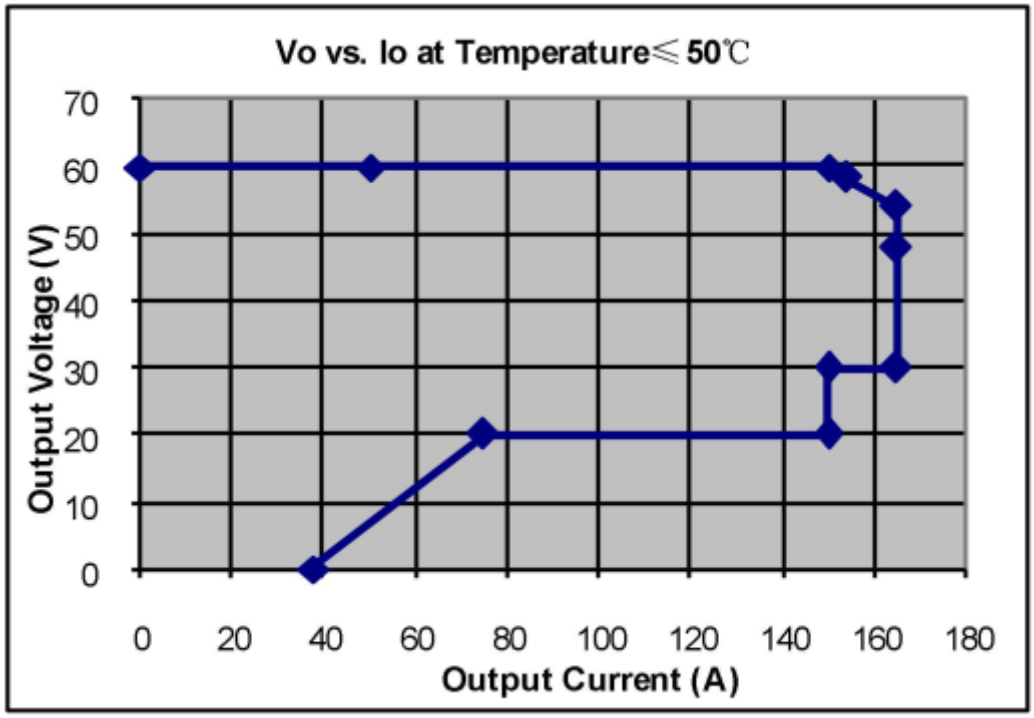


Figure 2.1 Output V-I Curve of BA48-9000

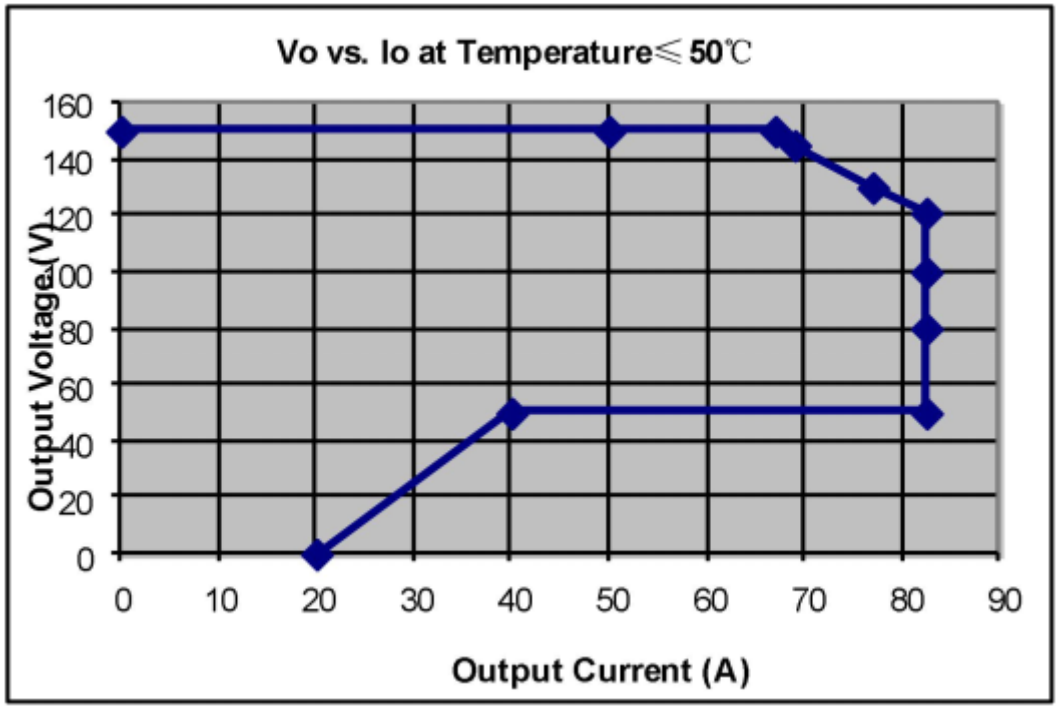


Figure 2.2 Output V-I Curve of BA150-10000

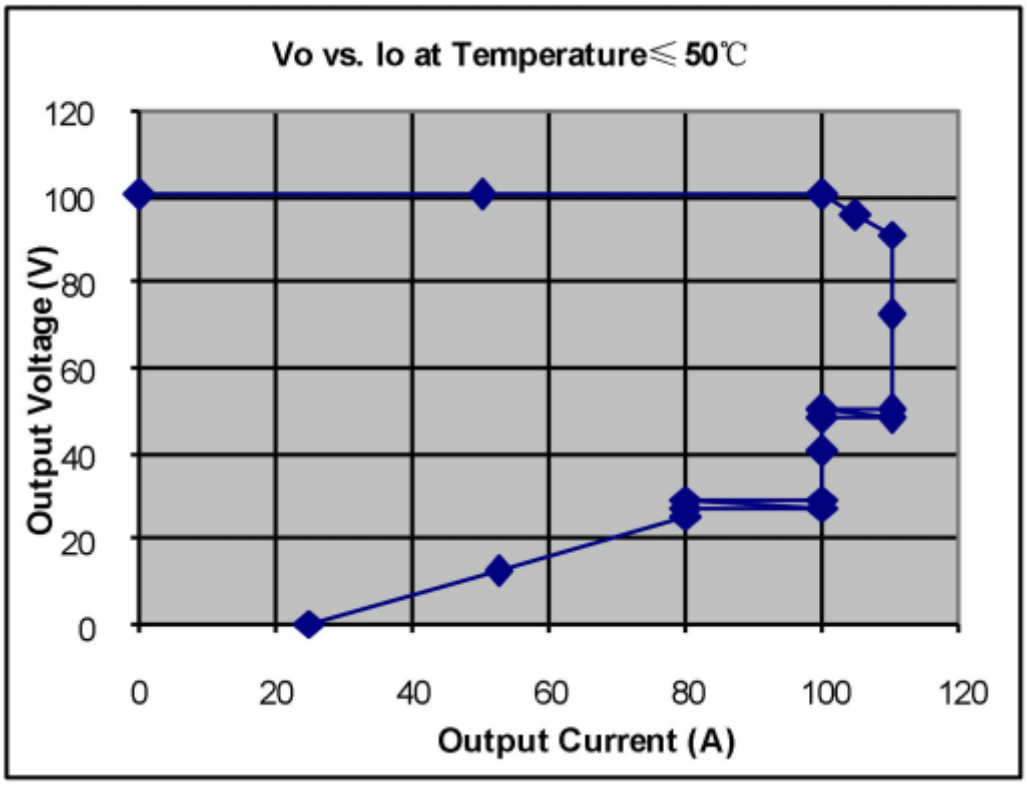
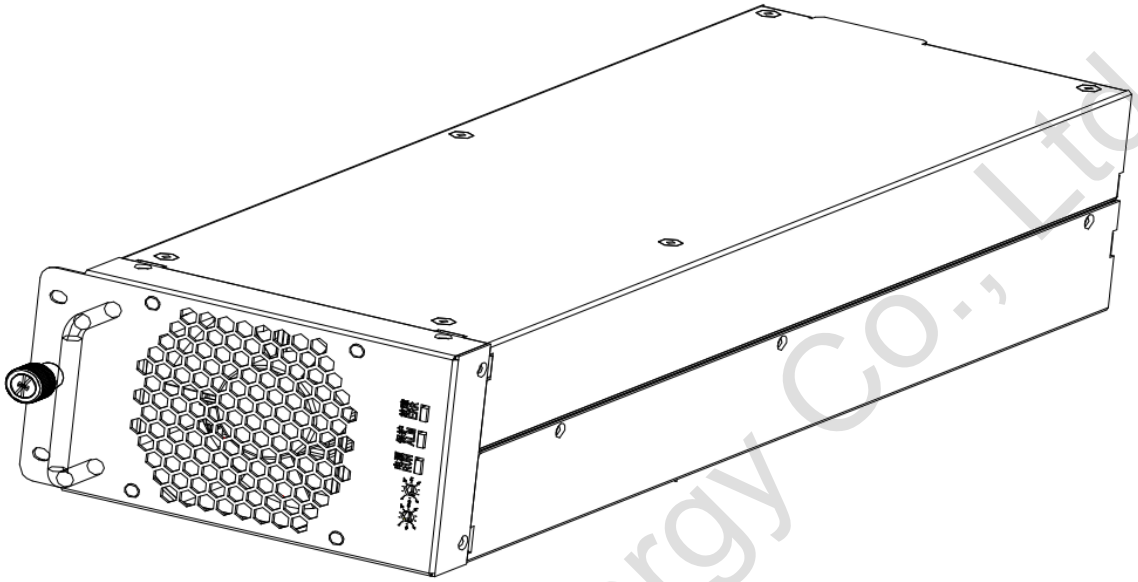


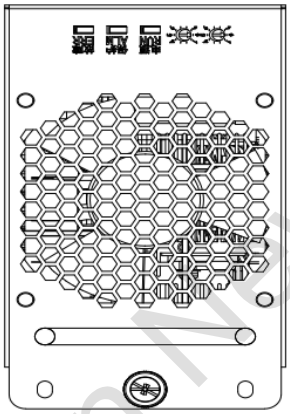
Figure 2.3 Output V-I Curve of BA100-10000

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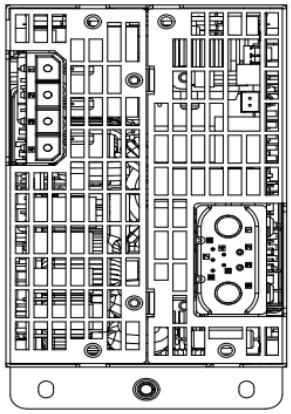
4 Appearance and Interface

4.1 Appearance





Front View



Rear View

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4.2 Input and Output interface

The various input and output signals of the BA100-10000 series ACDC module must be connected through the sockets at the back of the module. As shown in Figures 3 and 4, the definitions of each pin are as listed in Table 2:

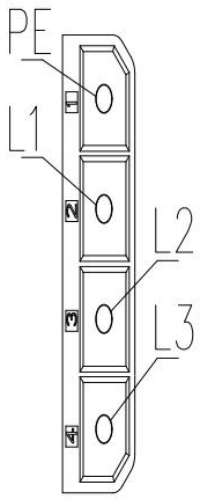


Figure 3: Input connector definition

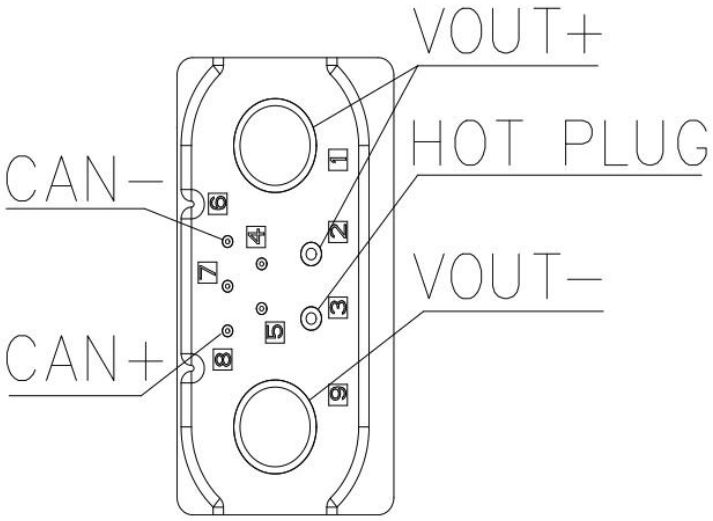


Figure 4: Output connector definition

Table 2: ACDC module socket Pin definition

Connector	Pin No.	Definition	Function Explain
DC INPUT	1	PE	The protective ground lead-out terminal of the module is connected to the module housing.
	2	One phase in three-phase electricity	The input terminal of the module's AC power supply adopts a three-phase three-wire connection method, and there is no requirement for phase sequence.
	3	One phase in three-phase electricity	
	4	One phase in three-phase electricity	
DC OUTPUT	1	OUT+	DC output positive
	2	OUT+	DC output positive
	3	Hot Plug	Hot swapping, connected to the negative terminal of the DC output
	6	CAN_L(-)	CANBus
	8	CAN_H(+)	
	9	Out-	DC output negative

Caution:

1. ensure safety, please make sure to correctly connect the protective ground PE terminal in the AC input to the earth.

2. ensure the reliability of the system, each module's three-phase AC input must be independently connected to the incoming circuit breaker.

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5 Operating Instructions

The address identification method of the ACDC module includes two modes: software automatic address identification (the default mode) and hardware address identification. The difference in appearance between the two is whether the ACDC module panel has a decimal address toggle switch.

Software address recognition: Sort according to the built-in electronic barcodes in the module. The smallest address of the module is 1, and the address size increases sequentially from there upwards.

The ACDC module panel for hardware address identification has two decimal address toggle switches and three LED working indicator lights.

(1) Module Address Setting Diagram

The size of the address setting must be between 1 and 60. It cannot be 0 or greater than 60. Otherwise, an address error will occur and the monitoring system will trigger an alarm.

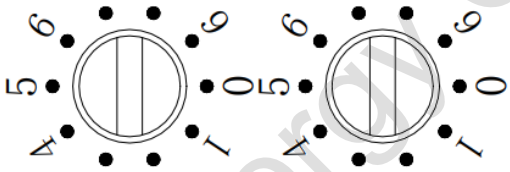


Figure 5: Input connector definition

(2) The LED indicator light alarm prompt of the module is as follows

- Green light on: AC input power supply indication.
- Green light flashes: Module identification, module enters sleep mode for standby.
- Yellow light on: Output current limit, high temperature power limit, AC low voltage power limit.
- Yellow light flashing: Module address duplication or module electronic barcode duplication;
- Red light on: Input under-voltage, input over-voltage, DC low voltage, internal over-temperature, internal bus under-voltage, internal bus over-voltage, environmental over-temperature, overcurrent protection, output over-voltage, fan failure.
- Red light flashing: fan failure.

Notice 1:

When there is a mixed insertion of software address modules and hardware address modules in the system, the N software address modules will automatically occupy addresses ranging from 1 to N, while the X hardware address modules need to set their addresses to (N+1) to (N+X).

Notice 2:

When protection occurs, the module will automatically shut off the output. The output overvoltage protection has a self-locking function. After confirming there is no fault, the

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power must be re-applied to restore it; other protections will be restored immediately after the fault is eliminated (while the module is powered on).

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6 Installation and Design

6.1 Module heat dissipation requirements

The module adopts a design of forward air intake and rear air exhaust for heat dissipation. When designing the system, it is necessary to ensure that there is at least 10 cm of intake space in front of the module panel, and also to guarantee a smooth intake channel; the exhaust port at the rear of the module ensures a smooth exhaust channel. The temperature rise at the exhaust port is 10°C compared to the intake port. Therefore, the rear system design of the module should try to avoid installing temperature-sensitive components.

6.2 Connector and cable assembly

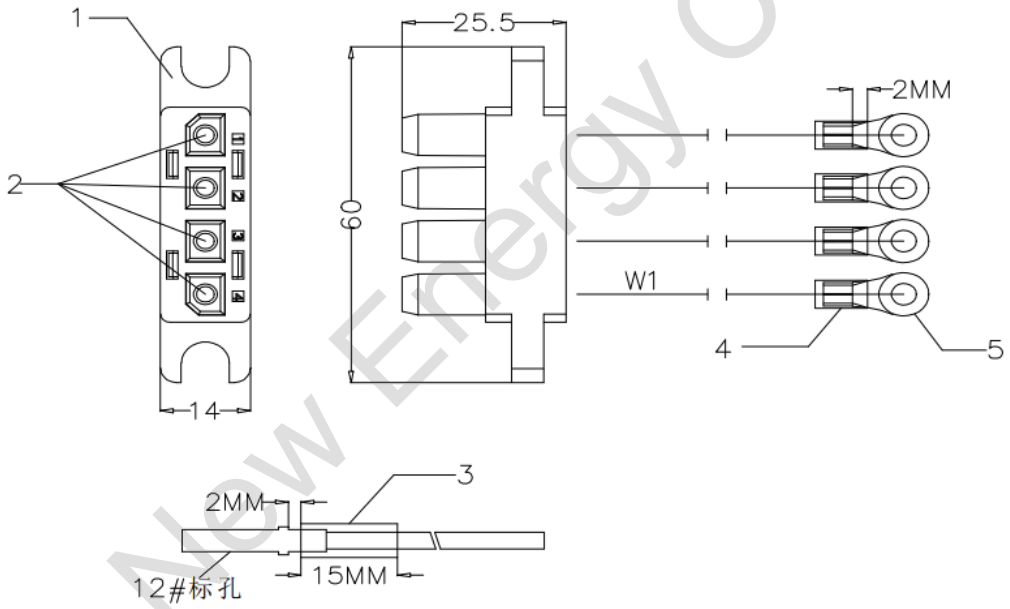


Figure 6: Input Connector Assembly

Part No.	Parts Name	Specification	Qty
W4	Cable	BVR-4mm^2 Red	1
W3	Cable	BVR-4mm^2 Green	1
W2	Cable	BVR-4mm^2 Yellow	1
W1	Cable	BVR-4mm^2 Yellow-Green	1
5	OT lug terminal	OT4-6	4
4	Shrink tube	φ8	4
3	Shrink tube	φ6	4
2	12# Pin terminal	JDC-TK-4Y	4
1	Connector socket	JDC-4YZ	1

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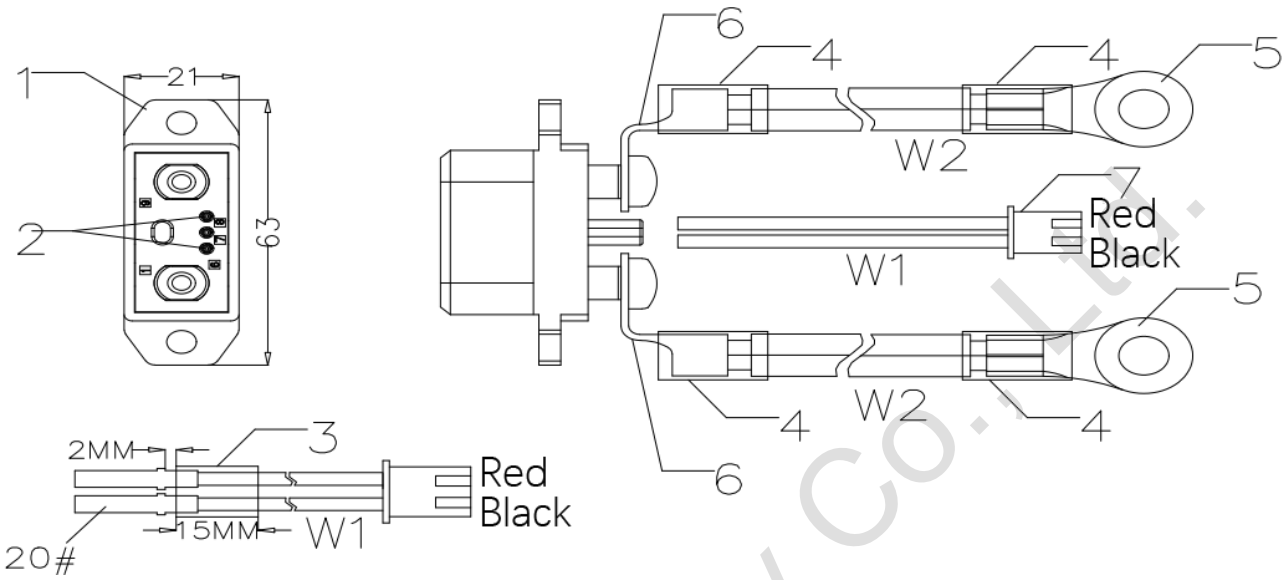


Figure 7: Output Connector Assembly

Part No.	Parts Name	Specification	Qty
W2	Cable	BVR-16mm^2 Yellow	2
W1	Cable	RVVP-300/300V 24AWG	1
7	XH2.54 connector	2.54mm distance	1
6	OT Lug terminal	90degree OT16-6	2
5	OT Lug terminal	OT16-6	2
4	Shrink tube	φ12 L=20mm	4
3	Shrink tube	φ8 L=15mm	1
2	20# Pin terminal	JDC-10Z-001Y	2
1	Connector socket	JDC-4YZ	1

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7 Dimension

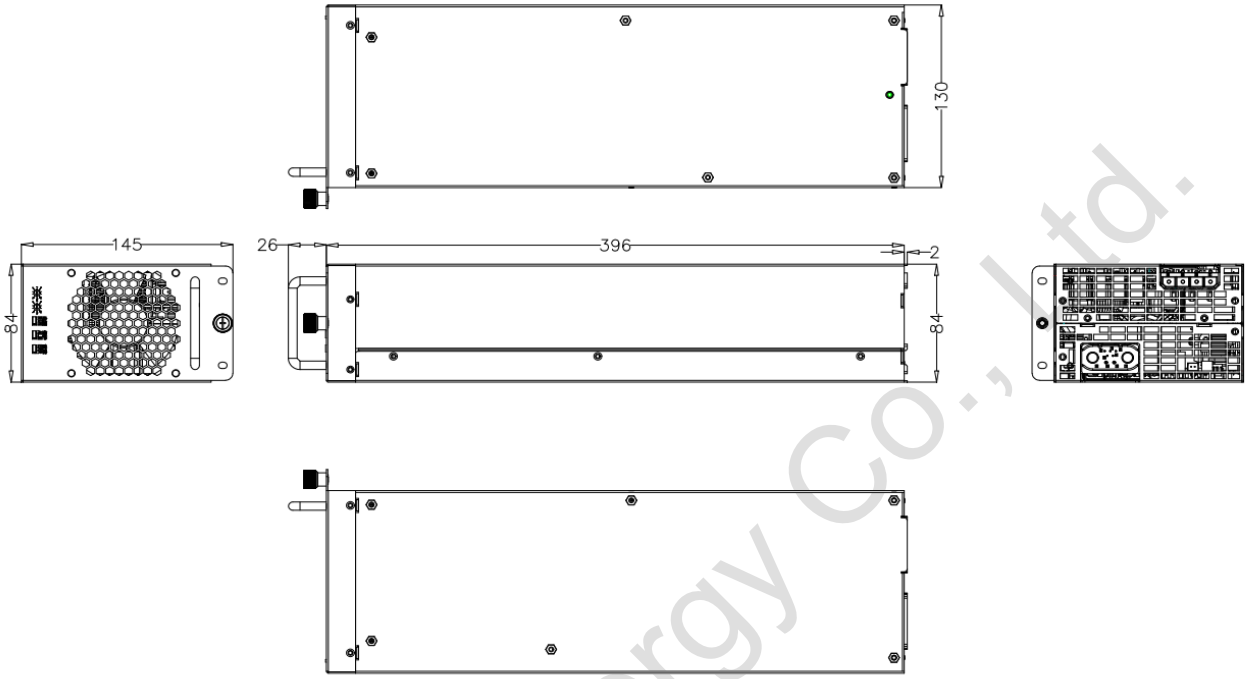


Figure 8: Module Outside Dimension (Unit mm)

8 Fault diagnosis

The ACDC module is equipped with a built-in CPU. When a fault occurs, the fault indicator light on the panel will illuminate. If it is determined that the ACDC module has a fault, please immediately loosen the fixing screws on the front panel of the ACDC module and remove the lower ACDC module. If there is a backup module, replace it with the backup module. This ACDC module supports hot swapping while powered on.

9 Packaging, transportation and storage

9.1 Packaging

This equipment is packed in cardboard boxes, individually. It is wrapped with shock-absorbing foam plastic.

Inside the packaging box, there is also a packing list, an installation and operation manual, and each rectifier module is labeled with a factory certificate of conformity.

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9.2 Transportation

The ACDC module transportation should be carried out in a packaged condition. During the transportation process, it must not be subjected to violent vibration or collision to prevent moisture absorption and rain exposure.

9.3 Storage

The ACDC module should be stored in an indoor environment that is at a temperature range of -40°C to +75°C, with a relative humidity not exceeding 95%, without corrosive gases, and with adequate air circulation. The storage period is one year.

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